

Instructor Answer Key

50 Multi-Select MCQs Based on Lab Sheets 06–08

Each question has exactly two or three correct options. Across the complete paper, each option letter A–E is correct exactly 25 times. The explanations below state why the selected combination is valid.

Question 01: A, E

The residual definition is actual minus predicted, and a zero residual is an exact fitted prediction. A positive residual indicates under-prediction, while MSE uses squared units and gives larger errors more influence.

Question 02: A, B, E

Scaling keeps feature magnitudes comparable, which matters to margin- and distance-based models. Fitting the scaler only on training data avoids information leakage; scaling cannot guarantee separation or change the task type.

Question 03: A, B

Random Forest combines bootstrap sampling with random feature selection. Sequential residual correction describes boosting, while identical data and split choices would reduce rather than create diversity.

Question 04: D, E

K-Means alternates between nearest-centroid assignment and mean-based centroid updates. It does not use class labels, linkage merging, or DBSCAN-style noise labels.

Question 05: A, B

Hyperparameters should be selected using training data and cross-validation, after which the test set provides one final estimate on unseen data. Repeated test use and preprocessing before the split leak information.

Question 06: B, C

Gradient descent subtracts a learning-rate-scaled gradient. The gradient supplies direction and the learning rate supplies step size; an excessively large rate can prevent convergence.

Question 07: A, B, C

DBSCAN is density based, can identify arbitrary shapes, and explicitly marks noise with label -1 . It does not require a preset K and does not use mean centroids as its defining mechanism.

Question 08: C, D

The separating boundary is the zero level set of the decision function, and its sign determines the binary prediction. The margins are at $f(x) = \pm 1$, and decision scores are not automatically probabilities.

Question 09: B, C

Agglomerative clustering is bottom-up: singleton clusters are successively merged. A dendrogram can be built before choosing a horizontal cut and therefore before fixing the final cluster count.

Question 10: A, E

MAE and RMSE are both expressed in the target's original units. MSE is in squared units, while R^2 is unitless.

Question 11: C, D, E

Gradient Boosting builds a sequence in which each learner corrects the current ensemble in the direction of reduced loss. It commonly uses weak, shallow trees rather than independent bootstrap trees.

Question 12: B, C, D

WCSS measures within-cluster squared variation and is minimized by K-Means. It decreases or stays unchanged as more clusters are allowed, while the elbow identifies diminishing improvement; label numbers themselves are arbitrary.

Question 13: C, D, E

Small C tolerates violations and often allows a wider margin, whereas large C emphasizes fitting the training points. Greater tolerance can reduce sensitivity to a suspected mislabelled observation.

Question 14: A, D, E

Voting and averaging are standard combination rules. Diversity can improve reliability when member errors differ, but improvement is not guaranteed and supervised ensembles still require targets.

Question 15: B, C, D

The tube defines a no-penalty region around the regression function. The loss is $\max(0, |y - f(x)| - \varepsilon)$, and observations on or beyond its boundary can constrain the solution.

Question 16: **D, E**

Adding independent trees often reduces prediction variability and improves stability, but accuracy need not increase monotonically and usually reaches a plateau.

Question 17: **A, B**

Standardization makes feature scales comparable and prevents variables such as income from dominating variables such as age. It changes distances and does not generate labels.

Question 18: **C, D**

Hinge loss is zero only when the signed margin is at least one. A correctly classified point inside the margin still has positive loss, and the maximum with zero prevents negative values.

Question 19: **A, E**

A useful clustering solution should be stable and interpretable in the application. Algorithmic convergence and arbitrary numeric IDs do not establish real-world truth.

Question 20: **D, E**

The Cartesian product contains eight combinations, and 5-fold cross-validation evaluates each across five train-validation partitions. The independent test set remains reserved for the final evaluation.

Question 21: **B, C, D**

The simple linear model contains an intercept and a slope, with the slope describing change per unit of the predictor. It predicts a continuous value rather than a sign-based class.

Question 22: **D, E**

Maximum depth controls individual-tree complexity. Greater depth can improve training fit but may capture noise and reduce generalization.

Question 23: **C, D**

Single linkage is based on the closest pair, while complete linkage is based on the farthest pair. Average linkage uses the average of cross-cluster pairwise distances.

Question 24: **A, D, E**

A larger ϵ creates a wider no-penalty tube. More observations may fall inside it, often leaving fewer support vectors, although the fitted line can still change.

Question 25: **C, D**

Randomly limiting the candidate features at splits is a defining Random Forest mechanism and affects tree correlation. No single setting is guaranteed to be best on every dataset.

Question 26: **A, D, E**

The coefficient of determination compares unexplained and total variation; one indicates zero residual error, while very poor out-of-sample predictions can make it negative. It is unitless and not confined to $[0, 1]$ on arbitrary test predictions.

Question 27: **B, C, D**

The test set simulates unseen data and should be used consistently across final comparisons. Repeated model decisions based on it create leakage and optimistic estimates.

Question 28: **A, B, C**

K-Means can converge to different local minima, so reproducible seeds and multiple starts are useful. Updated centroids are means and need not coincide with an observed point.

Question 29: **D, E**

The attribute summarizes split-based contribution, not causality or coefficient direction. Instability across runs can occur when correlated features can substitute for one another.

Question 30: **A, E**

The scaled correction is $0.1 \times 10 = 1$, so the updated prediction is $60 + 1 = 61$.

Question 31: **A, B**

Verification checks whether an established implementation produces similar parameters and predictions on the same regression problem. SVC solves a classification objective and is not the required verifier.

Question 32: C, D, E

The radius `eps` controls neighbourhood reach, while `min_samples` controls the local density requirement. Too small a radius fragments; too large a radius can merge structures.

Question 33: B, C

Support vectors are the influential boundary or violation points. Observations safely outside the margin have zero hinge loss and do not directly determine the fitted hyperplane.

Question 34: A, B, E

The learning rate is a shrinkage factor on each new learner. Smaller steps commonly require more estimators, whereas overly large steps can overshoot or overfit.

Question 35: B, C

Cluster numbers are arbitrary names. Two runs can represent identical partitions with different numeric labels, so memberships and centroid structure should be compared instead.

Question 36: C, D, E

Random Forest targets the high variance of individual trees by averaging diverse learners. If the trees were identical, their errors would remain highly correlated and aggregation would add little.

Question 37: B, C, D

The loss ignores error within the tube and penalizes only the excess beyond ε . The maximum with zero ensures that it is never negative.

Question 38: A, B, C

Fusion height records how dissimilar merged clusters were, and a horizontal cut converts the hierarchy into a selected partition. Linkage choice can change the complete merge sequence.

Question 39: A, D, E

Supervised learning learns from targets, while clustering searches for structure without them. Cluster interpretation requires later examination and should not use hidden generator labels as training targets.

Question 40: A, B, E

Random Forest uses independent diversified trees built from bootstrap samples and random feature subsets. Sequential correction and learning-rate shrinkage belong to Gradient Boosting.

Question 41: A, B

MSE squares each residual before averaging, so signs do not cancel and large errors receive greater weight. Its value is nonnegative and measured in squared target units.

Question 42: C, D, E

The K-Means update uses means because they minimize squared distances. Robust code handles empty clusters and stops on centroid stability or a maximum-iteration safeguard.

Question 43: A, B, C

The three stated settings control ensemble length, learner complexity, and correction size. Their effects interact, so larger values do not guarantee better generalization.

Question 44: A, E

The coordinate differences are 6 and 8, producing distance $\sqrt{36 + 64} = 10$. Large feature scales can dominate such a calculation, motivating standardization.

Question 45: A, B, C

The decision boundary is at zero and the two canonical margins are at plus and minus one, with total width $2/\|w\|$. Maximizing this width is the core geometric preference.

Question 46: A, B, E

Both are tree ensembles, but Random Forest aggregates independent diversified trees, while Gradient Boosting adds scaled sequential corrections and therefore uses a learning rate.

Question 47: B, C

DBSCAN's density connectivity can capture non-convex moon shapes and label noise. K-Means uses centroid proximity and specifying two centroids does not guarantee recovery of two curved groups.

Question 48: A, D, E

Overfitting appears as a generalization gap. Complexity controls and cross-validation help select settings without rewarding memorization or contaminating the test set.

Question 49: C, D

For a binary linear SVC, the stored coefficients and intercept reconstruct the decision function. The sign gives the class, but the raw score is not a calibrated probability.

Question 50: A, B, E

A useful unsupervised solution should be stable, interpretable, and practically meaningful. More clusters or algorithmic convergence alone does not establish a true grouping.

Balanced key check: 25 questions contain two correct options and 25 contain three. Each answer position A–E occurs exactly 25 times.